DERMAGENIE-AI SKINCARE ASSISTANT

AI SKIN INTELLIGENCE & PERSONALIZED SKINCARE PLATFORM

PROJECT REPORT

Submitted by

M.HARSHITHA

1.Introduction

Skincare is one of the fastest-growing consumer categories, yet most people still choose products by advertising, peer recommendations, or trial and error. Skin condition depends on many interacting factors: the underlying skin type, existing concerns such as acne, dryness, pigmentation or sensitivity, daily habits like hydration and diet, sleep quality, stress, physical activity, and exposure to sun and pollution. A routine that works for one person may irritate another, and few consumers have access to a dermatologist who can review all of these factors together. AI Skin Intelligence is a web-based platform that addresses this gap. It collects a user's skin profile, lifestyle habits, sleep patterns, and environmental exposure, then uses a combination of rule-based scoring, a trained machine learning model, and curated ingredient and product data to produce a personalized skincare plan. The platform also tracks progress over time, so that users and consultants can see whether the routine is working. The same system serves individual consumers as well as dermatology clinics, wellness platforms, beauty brands, and independent skincare consultants, who can review client data and consultation requests through dedicated dashboards.

2. Problem Statement

The skincare decision process currently suffers from four main problems. First, personalization is shallow. Most product finders ask one or two questions, such as skin type, and return a generic list. They rarely consider lifestyle factors, sleep, or stress, even though these influence skin condition measurably.

Second, ingredient knowledge is hard to apply. Consumers see long ingredient lists but cannot easily tell which ingredients suit their concern, which conflict with their allergies or sensitivity, or which combinations should not be used together.

Third, progress is rarely measured. Without a structured record of routine adherence and skin condition over time, users cannot tell whether a product is helping, and consultants have no objective data to guide follow-up advice. Fourth, professional access is limited. Clinics and consultants need a way to manage client profiles, assessments, and consultation requests, but generic tools do not support this workflow. The problem, therefore, is to build a system

that turns a person's full context into an explainable, personalized, and trackable skincare plan that both consumers and professionals can use.

3. Objectives

The project aims to build an AI-powered skincare intelligence platform with the following objectives. To collect a structured skin profile and lifestyle, sleep, and environmental data from each user. To produce an AI-driven skin assessment that calculates a skin health score, identifies and prioritizes skin concerns, and predicts the primary concern with a trained classifier. To generate personalized morning, evening, and weekly routines that match the user's skin type, concerns, and season. To analyse ingredients for suitability, allergy conflicts, and interactions, and to explain what each ingredient does. To recommend products within the user's budget, with alternatives and complete combinations. To track daily check-ins, routine adherence, and skin progress over time, including before-and after comparisons. To provide role-based dashboards for consumers, consultants, dermatologists, and administrators, along with consultation requests and exportable reports.

4. Approach

The project followed an iterative, milestone-based development approach. Each milestone delivered a working slice of the system, which was then extended in the next iteration. The work was organized into the following stages. Requirements analysis came first. The target users were identified as consumers, dermatology clinics, wellness platforms, beauty brands, and consultants, and their needs were translated into functional modules such as assessment, routine planning, and progress tracking.

Next, the data design was completed. A relational schema of fifteen tables was created to hold users, skin profiles, lifestyle and sleep logs, assessments, routines, adherence records, products, ingredients, ingredient concerns, daily check-ins, reminders, notifications, and consultation requests. The scoring and decision logic was then built as a set of rule-based services. These compute skin condition, lifestyle, sleep, routine consistency, and hydration scores, combine them into an overall skin health score, and rank concerns by severity. These rules are deliberately transparent so that each result can be explained to the user. A machine learning component was added next. A random forest classifier was trained on a skin concern dataset with categorical features such as skin type, sensitivity, hydration, stress, and sleep quality, along with

numeric age and water intake. The model is wrapped in a scikit-learn pipeline with one-hot encoding, so that preprocessing and prediction always run together. Its prediction is used as the primary concern when a clear result is available. Finally, the REST API and the frontend were developed and connected, and the system was tested, containerized, and documented.

5. Features

The platform provides the following modules, each implemented as a backend route group with a corresponding frontend page.

Module	Description
Authentication and roles	Registration, login with JWT tokens, password change, and role-based access for users, consultants, dermatologists, and administrators.
Skin profile	Captures skin type, sensitivity, concerns, allergies, and budget, and allows consultants to view patient profiles.
Lifestyle and sleep	Records daily habits such as hydration, physical activity, stress, and sleep duration and quality.
Skin assessment	Calculates skin health and component scores, ranks concerns by severity, identifies risk factors, and runs the ML concern prediction.
Personalized routine	Generates morning, evening, weekly treatment, seasonal, and adaptive routines based on skin type and primary concern.
Ingredient intelligence	Checks ingredient suitability against concerns and allergies, detects interactions between ingredients, and provides ingredient education.
Product recommendations	Ranks products against the profile and budget using scoring and cosine similarity, offers alternatives and complete combinations, and supports product comparison
Progress tracking, Routine adherence and check-ins	Shows score trends, improvement status, and before-and-after comparisons across assessments. Lets

	users log daily check-ins and routine completion, with summary statistics.
Health insights and analytics, Consultations	Produces health insights and dashboard analytics from lifestyle and sleep data. Allows users to request consultations, and lets consultants and dermatologists review and respond.
Notifications and reminders	Creates reminders for routines, and manages read and unread notifications.
Reports and export	Exports assessment, routine, product, progress, and health reports as PDF and Excel files.

6. Technology Stack

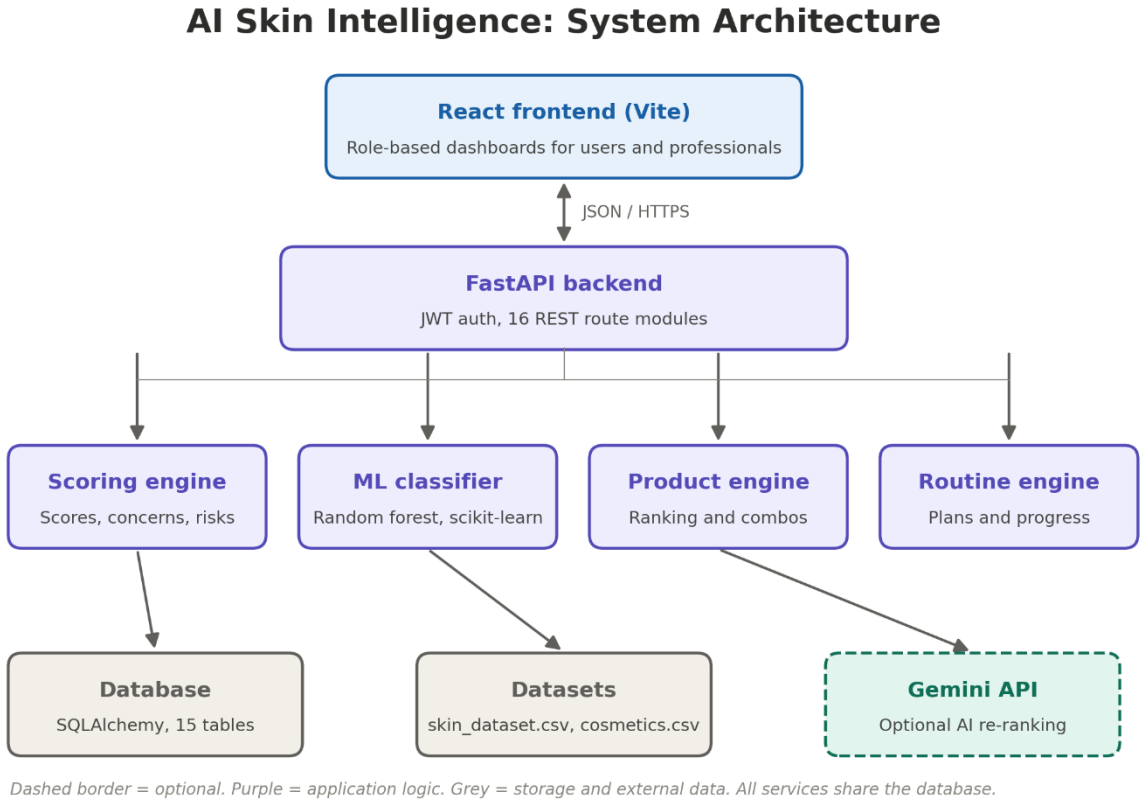
The platform is split into a React frontend and a Python backend, connected through a JSON REST API.

Layer	Technology	Purpose
Frontend framework	React 19 with React Router 7	Single-page application with protected, role-based routes
Build tool	Vite 8	Development server and production build
Charts	Recharts 3	Progress, score, and analytics visualizations
Backend framework	FastAPI with Uvicorn	REST API with automatic request validation and documentation
Data validation	Pydantic 2	Request and response schemas
Database access	SQLAlchemy 2.0	ORM for all relational models
Database	SQLite (development), PostgreSQL (shared deployment)	Persistent storage, selected through an environment variable
Authentication	JWT (python-jose), bcrypt	Stateless login tokens and secure password hashing
Machine learning	scikit-learn random forest, pandas, joblib	Skin concern classification and model persistence

Layer	Technology	Purpose
Generative AI (optional)	Google Gemini via the google-genai SDK	Product re-ranking and ingredient explanations
Report generation	ReportLab (PDF), openpyxl (Excel)	Downloadable reports
Containerization	Docker, nginx	Packaging backend and frontend for deployment

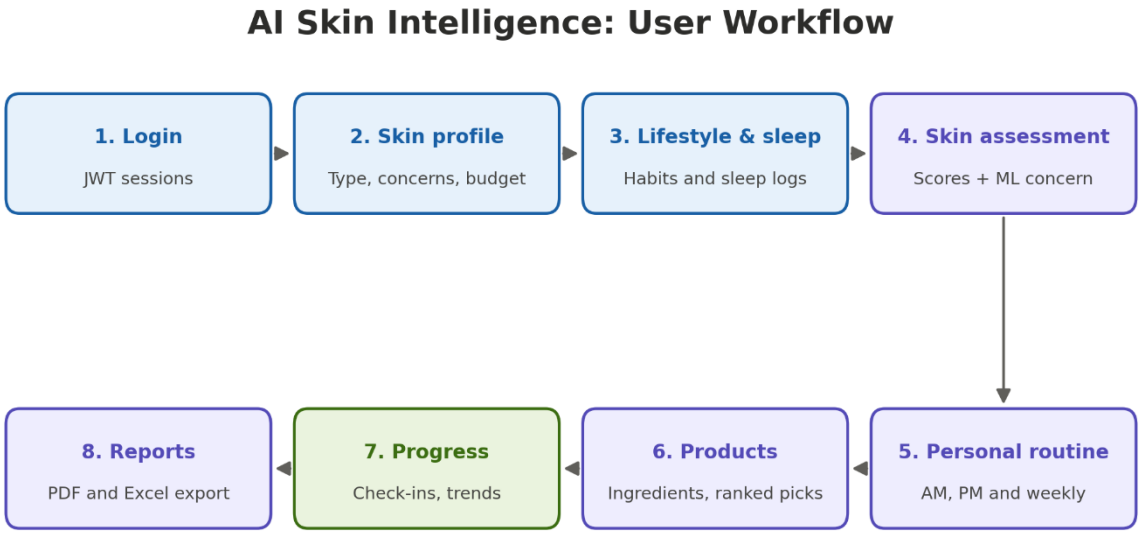
7. System Workflow and Architecture

The system follows a layered architecture. The React frontend sends authenticated requests to the FastAPI backend. The backend routes delegate work to service modules, which implement scoring, machine learning prediction, ingredient analysis, product ranking, routine generation, and progress analysis. All services read from and write to the relational database through SQLAlchemy. The Gemini API is called only from the product and ingredient services, and only when it is configured.



A typical user journey follows a clear sequence. The user registers or logs in and completes the skin profile and lifestyle and sleep logs. The user then runs a skin assessment, which stores the scores, concerns, and predicted concern. From the

assessment, the system generates a personalized routine, recommends products that match the routine and budget, and checks each ingredient for suitability. The user then performs daily check-ins and routine adherence logs. Progress analysis compares assessments over time, and reports can be exported at any point. Consultants and dermatologists follow a parallel path, in which they view patient profiles and assessments and respond to consultation requests.



8. Deployment

The AI-Powered Skin Intelligence and Personalized Skincare Routine Recommendation Platform was successfully deployed on AWS EC2 using Docker and Docker Compose. The FastAPI backend and React frontend are running in separate containers, with Nginx serving the frontend and forwarding API requests to the backend. SQLite data is stored in a persistent Docker volume to maintain database information across container rebuilds.

The application was made publicly accessible using a free DuckDNS domain, skin-ai.duckdns.org, and secured with a free SSL/TLS certificate from Let's Encrypt. Nginx was configured to redirect HTTP traffic to HTTPS, and successful tests confirmed that the secure website returns an HTTP 200 response. Automatic certificate renewal was configured using the Amazon Linux 2023 cron scheduler, with a successful renewal simulation.

Overall, the deployment establishes a publicly accessible, containerized application with encrypted communication, persistent database storage, and a

scheduled certificate-renewal process. **The platform is live and accessible at https://skin-ai.duckdns.org.** Periodic monitoring of application logs, database backups, and certificate-renewal logs is recommended to maintain reliability and security.

[**link :https://skin-ai.duckdns.org**]

9. Results

The platform meets its functional objectives. A user can progress from registration to a complete assessment, a personalized routine, product and ingredient recommendations, and tracked progress, and can export the results as PDF or Excel. Consultants and dermatologists can view client data and respond to consultation requests through their own dashboards, and administrators can manage users.

The assessment engine produces scores and concern rankings that are explainable, because each score is derived from named inputs and each concern has a severity value. The machine learning model adds a data-driven prediction of the primary concern, and the platform records its confidence so that low-confidence predictions can be recognized.

The recommendation layer returns budget-aware products, alternatives for each product, and combinations that work together. Ingredient checks flag allergy conflicts and known interactions, which gives users a clear reason for each recommendation.

10. Conclusion

AI Skin Intelligence demonstrates that personalized skincare guidance can be built from a combination of transparent rules, a trained machine learning model, and curated product and ingredient data, with generative AI as an optional enhancement. The platform brings together assessment, routine planning, ingredient analysis, product recommendation, and progress tracking in one system, and it serves both individual consumers and professionals who need to manage client skin care.

The project also shows the value of a layered design. Because the scoring, machine learning, and recommendation logic are separated into service modules behind a stable API, each part can be improved independently, and the system continues to work when the optional AI service is unavailable.

11. Future Improvements

Several extensions would make the platform more accurate and more useful. The first is to replace the current dataset with real, consented clinical data and to retrain and validate the model with a larger and more diverse sample, including different skin tones, ages, and climates. Image-based assessment is a natural next step: a camera-based analysis of skin texture and visible concerns would make the assessment richer and reduce the need for manual input, though it would require careful validation and privacy safeguards.

The second improvement is to complete the product recommendation module with a richer product catalogue and live pricing, and to add the planned product-matching step that uses the trained model's output more directly.

A third improvement is to add mobile applications and push notifications so that routine reminders and check-ins reach users in daily life.

Fourth, the progress module could include longer-term analytics such as seasonal trends and correlation between lifestyle changes and skin scores. Fifth, the consultant tools could be extended with secure messaging, appointment scheduling, and clinic-level reporting for dermatology practices and beauty brands. Finally, the platform should undergo formal security review and compliance checks, including data protection requirements for health-related information, before it is offered to the public.